

Recent Advancements in End-to-End Autonomous Driving using Deep Learning: A Survey

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ABSTRACT

End-to-End driving is a promising paradigm as it circumvents the drawbacks associated with modular systems, such as their overwhelming complexity and propensity for error propagation. Autonomous driving transcends conventional traffic patterns by proactively recognizing critical events in advance, ensuring passengers safety and providing them with comfortable transportation, particularly in highly stochastic and variable traffic settings. This paper presents a comprehensive review of the End-to-End autonomous driving stack. It provides a taxonomy of automated driving tasks wherein neural networks have been employed in an End-to-End manner, encompassing the entire driving process from perception to control. Recent developments in End-to-End autonomous driving are analyzed, and research is categorized based on underlying principles, methodologies, and core functionality. These categories encompass sensorial input, main and auxiliary output, learning approaches ranging from imitation to reinforcement learning, and model evaluation techniques. The survey incorporates a detailed discussion of the explainability and safety aspects. Furthermore, it assesses the state-of-the-art, identifies challenges, and explores future possibilities. We maintain the latest advancements and their corresponding open-source implementations at this [link](#).

1. Introduction

Autonomous driving refers to the capability of a vehicle to drive partly or entirely without human intervention. The modular architecture [1, 2, 3, 4, 5] is a widely used approach in autonomous driving systems, which divides the driving pipeline into discrete sub-tasks. This architecture relies on individual sensors and algorithms to process data and generate control outputs. It encompasses interconnected modules, including perception, planning, and control. However, the modular architecture has certain drawbacks that impede further advancements in autonomous driving (AD). One significant limitation is its susceptibility to error propagation. For instance, errors in the perception module of a self-driving vehicle, such as misclassification, can propagate to subsequent planning and control modules, potentially leading to unsafe behaviors. Additionally, the complexity of managing interconnected modules and the computational inefficiency of processing data at each stage pose additional challenges associated with the modular approach. To address these shortcomings, an alternative approach called End-to-End driving [6, 7, 8, 9, 10, 11] has emerged. This approach aims to overcome the limitations of the modular architecture.

The End-to-End approach streamlines the system, improving efficiency and robustness by directly mapping sensory input to control outputs. The benefits of End-to-End autonomous driving have garnered significant attention in the research community as shown in Fig. 1. Firstly, End-to-End driving addresses the issue of error propagation, as it involves a single learning task pipeline [12, 13] that

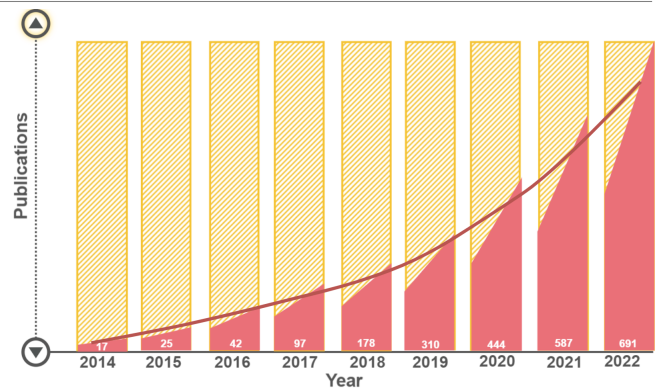


Figure 1: The number of articles in the Web of Science database containing the keywords 'End-to-End' and 'Autonomous Driving' from 2014 to 2022 illustrates the increasing trend in the research community.

learns task-specific features, thereby reducing the likelihood of error propagation. Secondly, End-to-End driving offers computational advantages. Modular pipelines often entail redundant computations, as each module is trained for task-specific outputs [4, 5]. This results in unnecessary and prolonged computation. In contrast, End-to-End driving focuses on the specific task of generating the control signal, reducing the need for unnecessary computations and streamlining the overall process. End-to-End models were previously regarded as "black boxes", lacking transparency. However, recent methodologies have improved interpretability in End-to-End models by generating auxiliary outputs [7, 13], attention maps [14, 15, 16, 9, 17, 18], and interpretable maps [18, 19, 8, 20, 12, 21]. This enhanced interpretability provides insights into the root causes of

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